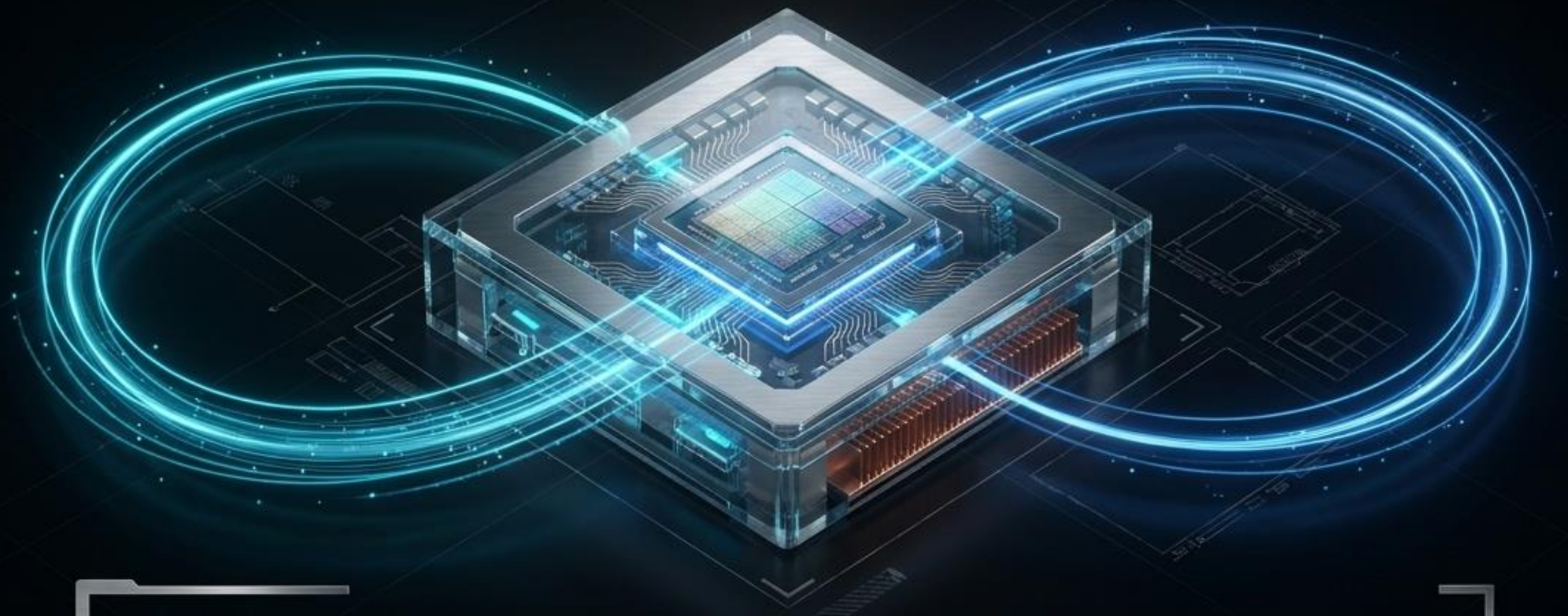
突破邊緣 AI 散熱極限

強化學習（RL）驅動的動態能耗管理



通往 2026 實體 AI（Physical AI）時代的關鍵調度技術

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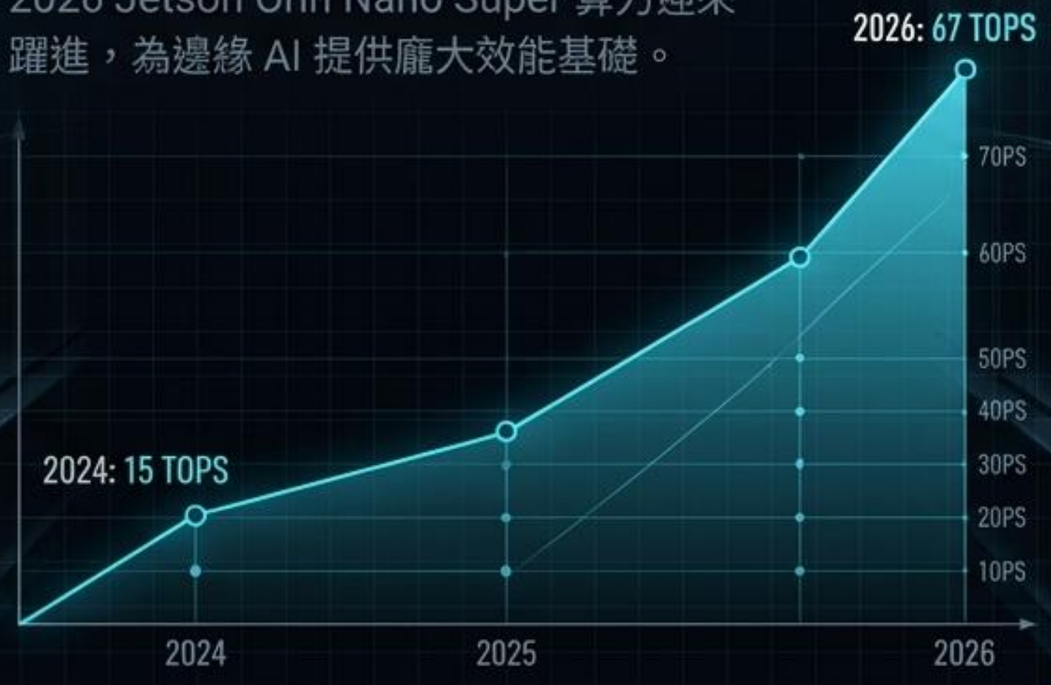
➢ Introduction

67 TOPS

算力空前激增

2026 Jetson Orin Nano Super 算力迎來躍進，為邊緣 AI 提供龐大效能基礎。



嚴峻的挑戰

散熱與續航的物理極限

傳統散熱已達物理極限，設備過熱導致的「強制降頻」成為阻礙大型模型落地的最大死穴。

續航力受限 (Battery Life Constrained)
CRITICAL LOW BATTERY

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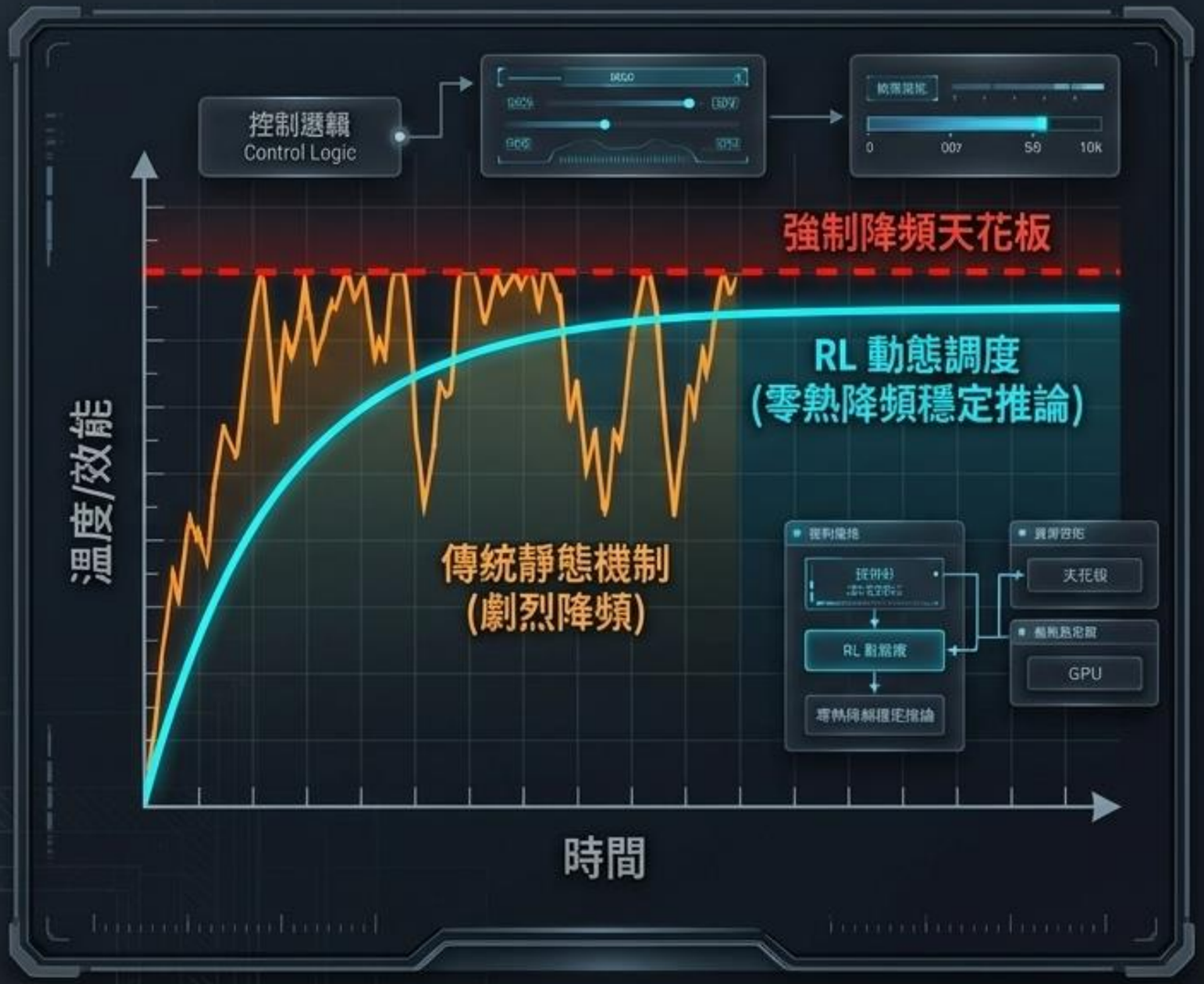
核心突破：實現零熱降頻



CPU-GPU 聯合優化引擎

捨棄傳統靜態預設規則，賦予設備自主思考與動態優化能力，在毫秒間精準分配運算權重。

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效能躍進：實機驗證數據

RL-DVFS 在邊緣運算平台上的決定性優勢

推論能耗大幅減少

透過精準分配電壓與頻率，有效阻斷不必要的能源消耗。

P95 延遲死守極限

在極致節能狀態下，依然確保即時反應與嚴格的延遲控制。

熱降頻次數 (Zero Throttling)

從根本上消除過熱導致的效能波動，實現絕對穩定的持續推論。

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➢ Related Work

多場景泛化驗證矩陣

單任務極致節能

2025 NCKU 碩論實測（Jetson Nano）

完美平衡能耗與效能

多任務高效泛化

2025 arXiv 報告（隨機負載測試）

具備高度適應性與輕量化優勢

實質節能效益驗證：無論負載型態為何，RL-DVFS 皆能穩定貢獻 3%~18% 的泛化節能表現。

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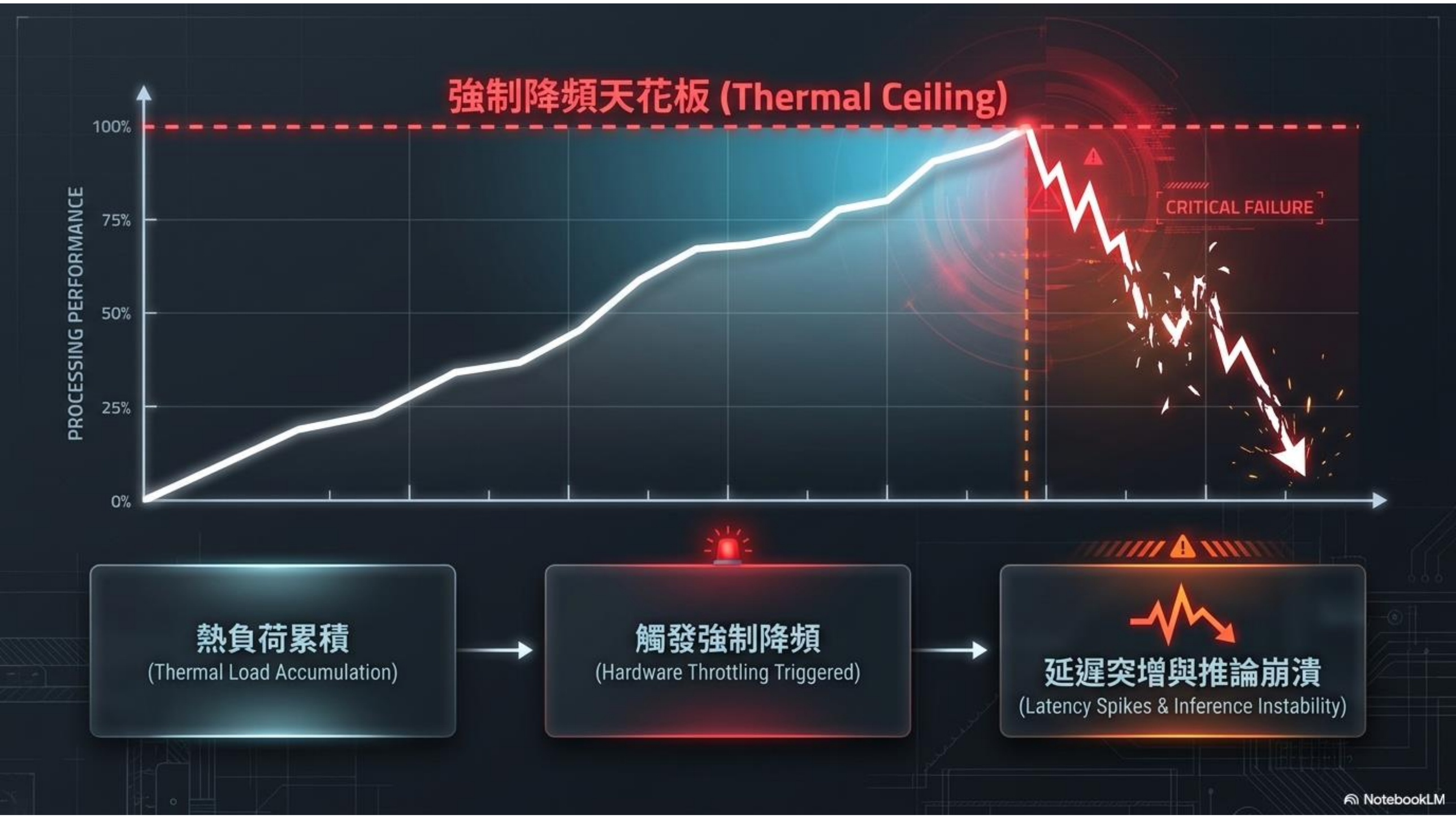
強制降頻天花板 (Thermal Ceiling)
PROCESSING PERFORMANCE
100%
75%
50%
25%
0%
CRITICAL FAILURE
熱負荷累積
(Thermal Load Accumulation)
觸發強制降頻
(Hardware Throttling Triggered)
延遲突增與推論崩潰
(Latency Spikes & Inference Instability)
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	傳統靜態 DVFS	RL-DVFS
控制邏輯 Control Logic	被動應付 (Reactive)	動態學習預判 (Predictive)
天花板處理 Ceiling Handling	觸頂強制降頻 (Hard Throttling)	零熱降頻預判 (Zero Throttling)
資源分配 Resource Allocation	獨立運作 (Isolated)	CPU-GPU 協同優化 (Joint Optimization)
能耗表現 Energy Profile	邊際效益遞減的高浪費 (High Waste)	精準電壓頻率分配 (Precision Allocation)

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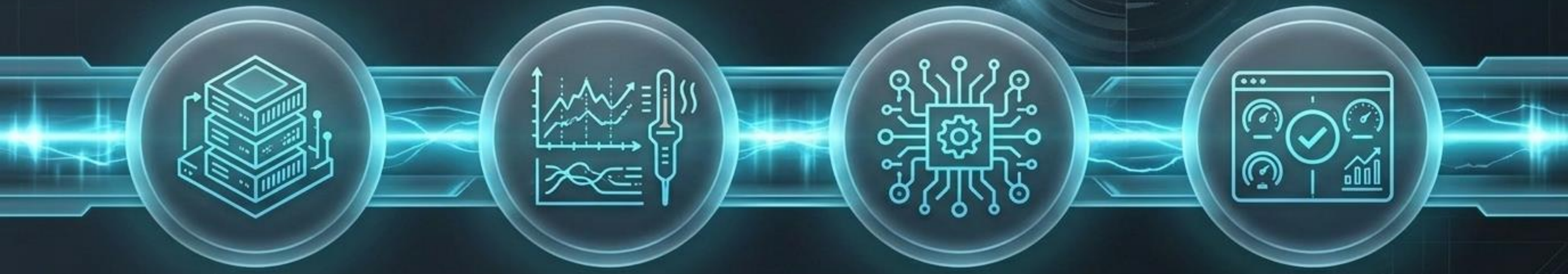
➢ Proposed Design



Phase 1:
環境架構建置
(Foundation)

Phase 2:
數據採集與熱特徵建模
(Observation)

Phase 3:
RL 代理程式開發
(Intelligence)

Phase 4:
實驗驗證與對比
(Validation)

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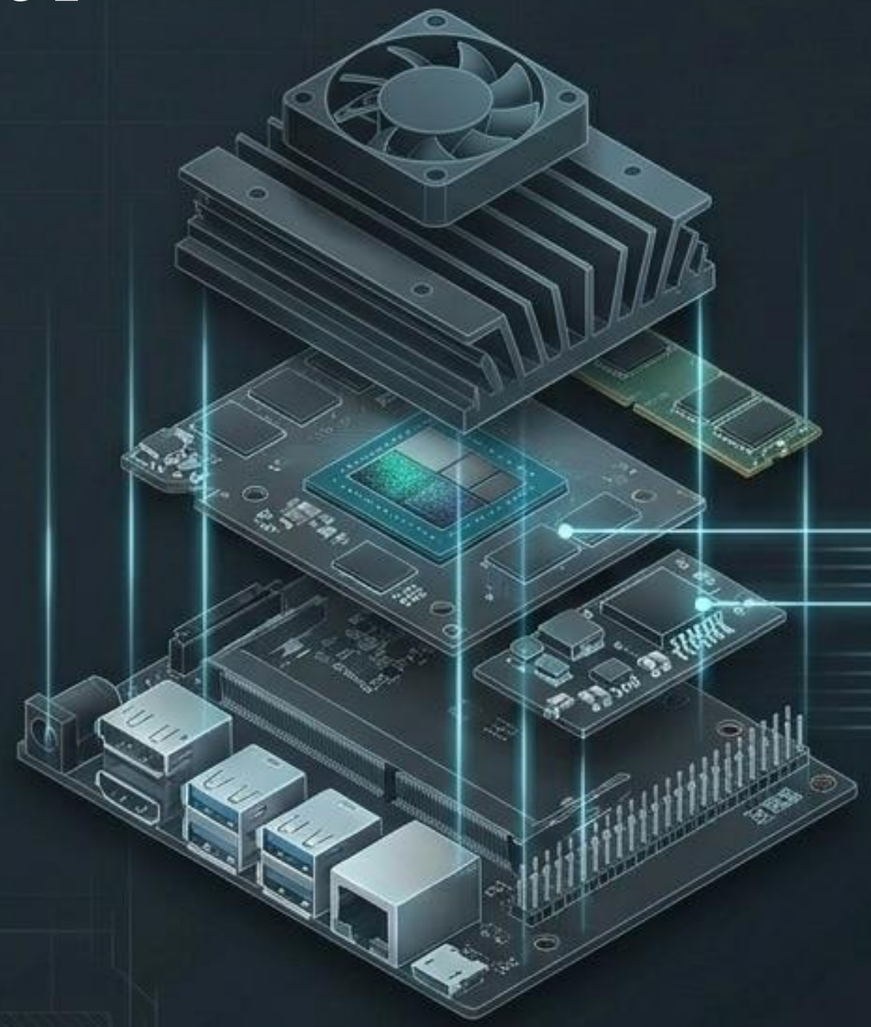
➢ Phase 1



平台核心：Jetson Nano 次世代架構
(Platform Core: Next-Gen Architecture)

底層工具鏈 - Native Toolchain

Panel A
tegrastats API
硬體狀態即時遙測
(Real-time Hardware Telemetry)

Panel B
nvpmodel 配置
底層功耗模式與頻率控制
(Power Mode & P-State Frequency Control)

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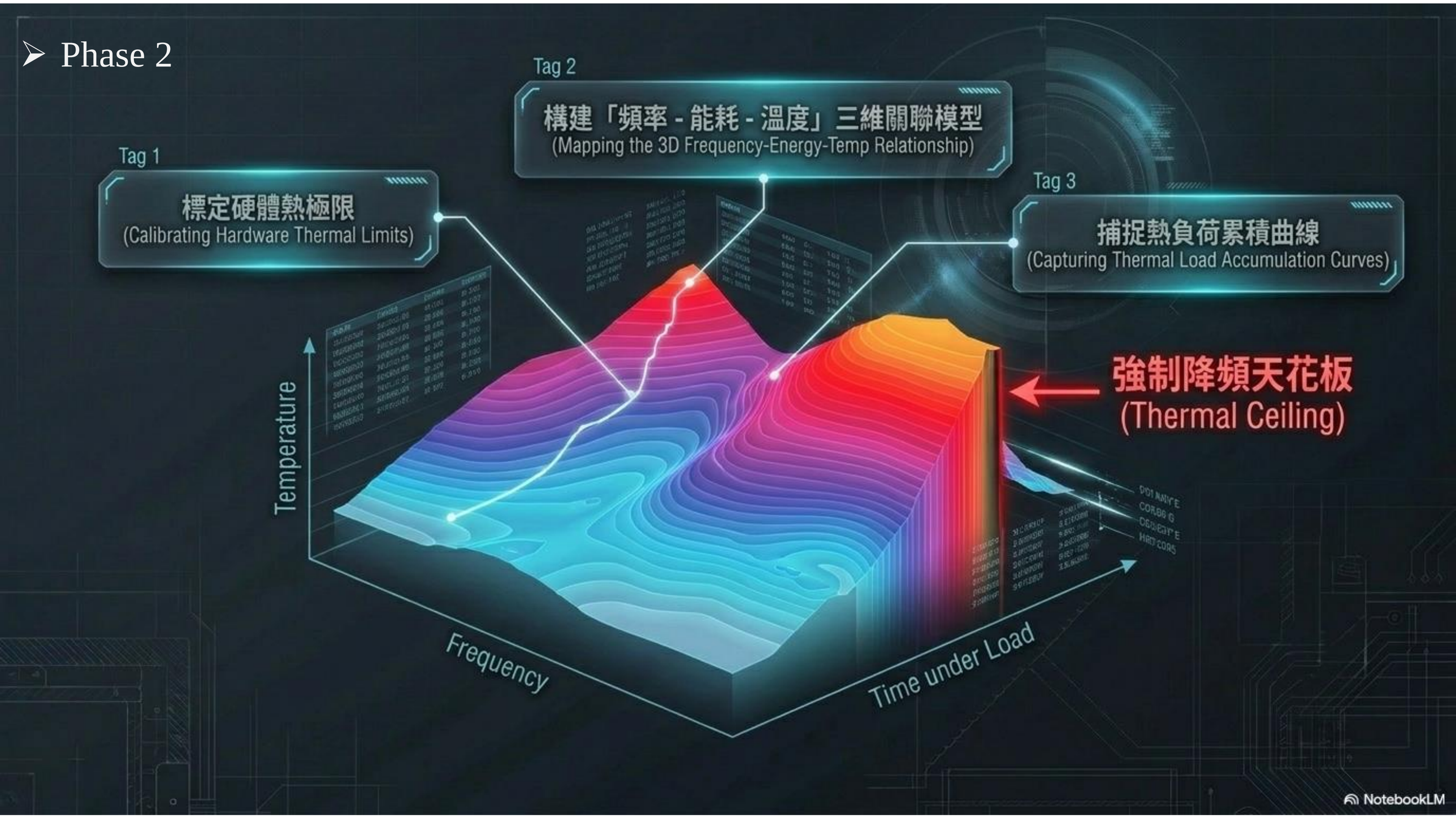
➢ Phase 2



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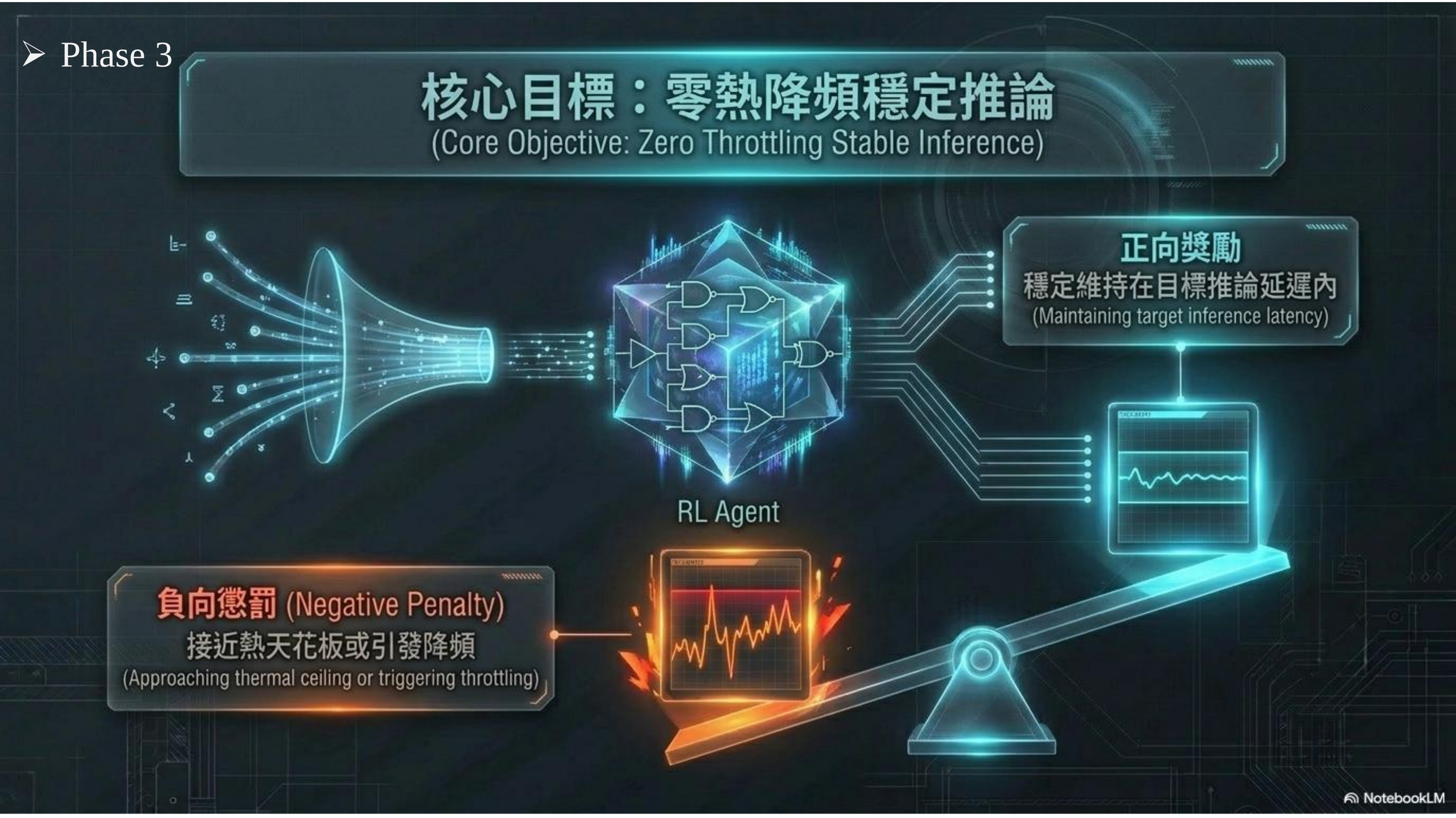
➢ Phase 3


核心目標：零熱降頻穩定推論
(Core Objective: Zero Throttling Stable Inference)
正向獎勵
穩定維持在目標推論延遲內
(Maintaining target inference latency)
RL Agent
負向懲罰 (Negative Penalty)
接近熱天花板或引發降頻
(Approaching thermal ceiling or triggering throttling)
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➢ Phase 3

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核心溫度
(Core Temperatures - CPU/GPU/Thermal Zone)
CPU 78°C
GPU 82°C
Zone 75°C
瞬時功耗
(Instantaneous Power Draw)
Total 150W
CPU 80W
GPU 70W
資源刻度
(Utilization % & Memory State)
85%
90%
CPU 85%
Memory 16GB / 24GB
微秒級系統感知：為代理程式提供絕對的環境解析度
(Microsecond System Perception: Providing the agent with absolute environmental resolution)
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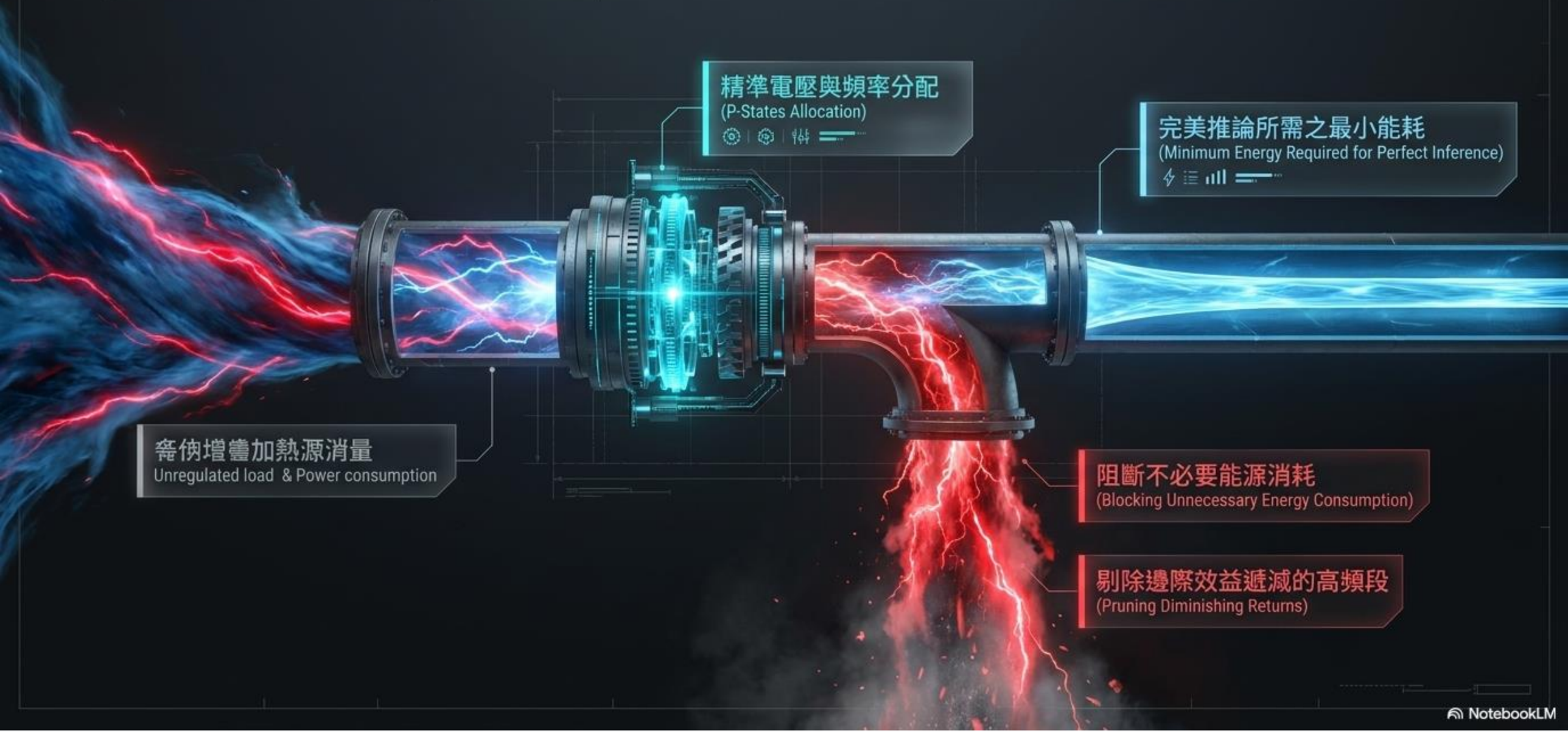
精準電壓與內機繕橫

Unregulated raw thermal load and power consumption



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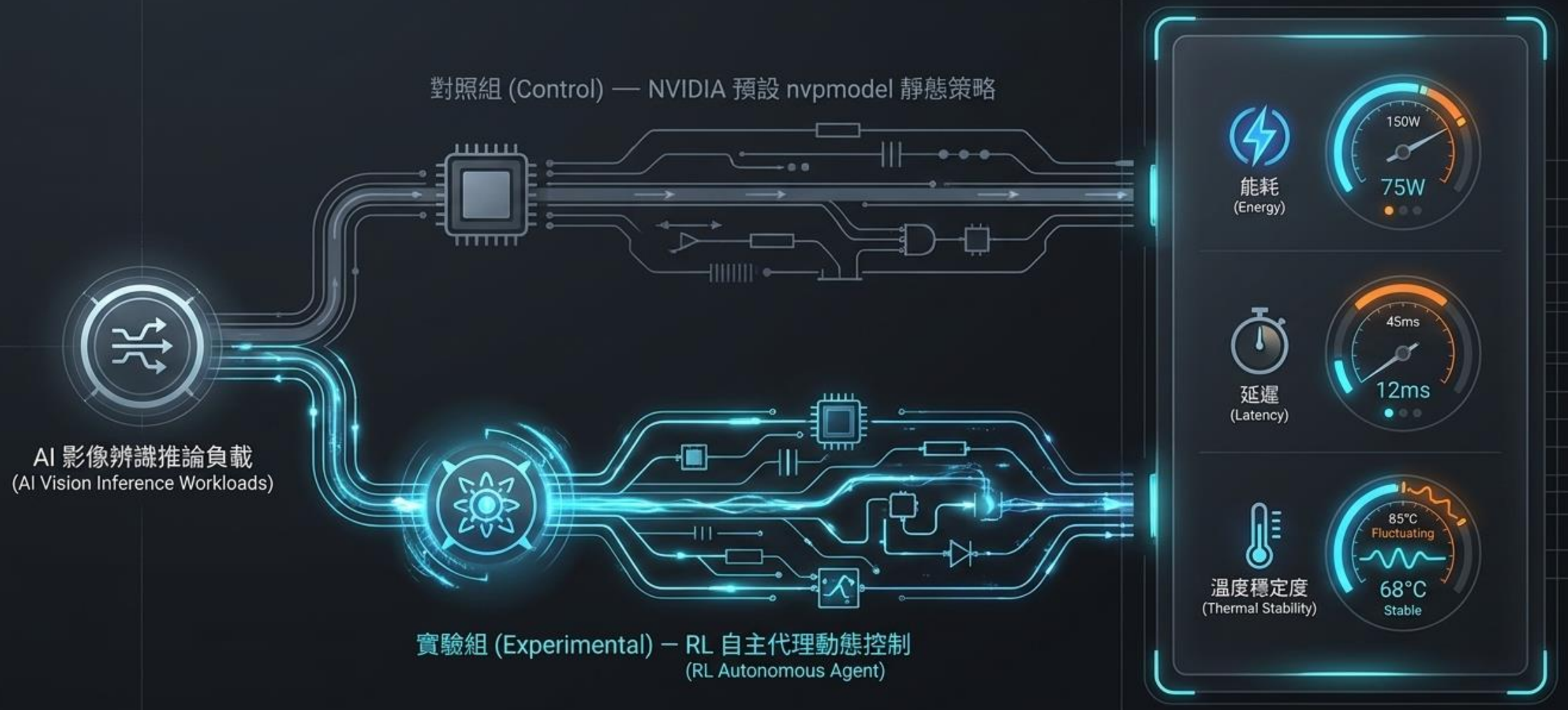
➢ Phase 4

The Precision Thermal Engine



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➢ 對比成效

- 18%

總體能耗降低

(Total Energy Consumption Reduced)

< 24ms

穩定推論延遲

(Stable Inference Latency Maintained)

100%

零熱降頻達成率

(Zero Throttling Success Rate)

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➢ 智慧能耗管理閉環

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